



Clinical Biochemistry

MBBS BIOCHEMISTRY · PAPER 1

Acid-Base Balance

Blood Buffers

Henderson-Hasselbalch

Liver Function Tests

Renal Function Tests

Jaundice

Free Radicals & ROS

Antioxidants

Xenobiotics & Cytochrome P450

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01 Acid-Base Balance

⚡ Normal blood pH = **7.35–7.45**. Maintained by three systems: (1) Buffer systems (immediate, seconds), (2) Respiratory compensation (minutes–hours), (3) Renal compensation (hours–days, most powerful).

NORMAL ARTERIAL BLOOD GAS (ABG) VALUES

PYQ 2014-S, 2018-S, 2023

pH 7.35–7.45

pCO₂ 35–45 mmHg

pO₂ 80–100 mmHg

HCO₃⁻ 22–26 mEq/L

Base Excess ±2 mEq/L

O₂ Saturation 95–100 %

► ACID-BASE REGULATORY SYSTEMS – HIERARCHY

IMMEDIATE (seconds) — Chemical Buffers $\text{HCO}_3^-/\text{H}_2\text{CO}_3$ · Phosphate ($\text{HPO}_4^{2-}/\text{H}_2\text{PO}_4^-$) · Protein/Haemoglobin · Ammonia buffer in kidney First line of defence — do NOT eliminate acid, just neutralise it temporarily

↓ If insufficient → Respiratory system kicks in

FAST (minutes to hours) — Respiratory Compensation $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3 \rightleftharpoons \text{H}^+ + \text{HCO}_3^-$ (catalysed by carbonic anhydrase)
 Acidosis → ↑RR → blow off CO_2 → pH rises
 Alkalosis → ↓RR → retain CO_2 → pH falls
 Lungs eliminate CO_2 (volatile acid). Fast but not complete correction.

↓ If persistent → Renal system — most powerful but slowest

SLOW but COMPLETE (hours to days) — Renal Compensation **Acidosis:** Kidneys ↑ H^+ excretion (as NH_4^+ and titratable acid) + ↑ HCO_3^- reabsorption
Alkalosis: Kidneys ↓ H^+ excretion + ↓ HCO_3^- reabsorption + ↑ HCO_3^- excretion
 Only system that can completely restore normal pH. Carbonic anhydrase inhibitors (acetazolamide) block renal H^+ secretion.

RENAL MECHANISMS FOR H^+ REGULATION

PYQ 2019-S, 2025

 H^+ Secretion (Acidosis — increase all three):

- ◆ **Titratable acidity:** $\text{H}^+ + \text{HPO}_4^{2-} \rightarrow \text{H}_2\text{PO}_4^-$ (phosphate buffer) — excreted in urine
- ◆ **Ammonium excretion:** Glutamine → Glutamate + NH_3 (in tubular cells) → $\text{NH}_3 + \text{H}^+ \rightarrow \text{NH}_4^+$ (excreted in urine). Most important during chronic acidosis — can increase 10-fold
- ◆ **HCO_3^- reabsorption:** All filtered HCO_3^- reabsorbed (mainly proximal tubule) — mediated by carbonic anhydrase

Key Renal Facts for Exam:

- ◆ Proximal tubule: reabsorbs 80–85% of filtered HCO_3^-
- ◆ Distal tubule + collecting duct: fine-tune H^+ secretion, NH_4^+ production
- ◆ Urine pH can range from **4.5 to 8.5** (wide range = renal power)
- ◆ Aldosterone → ↑ H^+ secretion in collecting duct (explains metabolic alkalosis in hyperaldosteronism)
- ◆ Carbonic anhydrase inhibitors (acetazolamide) → ↓ HCO_3^- reabsorption → metabolic acidosis

02

Blood Buffers & Henderson-Hasselbalch

HENDERSON-HASSELBALCH EQUATION

★ PYQ 2013-S, 2019-S, 2021

$$\text{pH} = \text{pK}_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

For bicarbonate buffer:

$$\text{pH} = 6.1 + \log [\text{HCO}_3^-] / [\text{H}_2\text{CO}_3]$$

$$\text{Normal ratio} = 20:1 \rightarrow \text{pH} = 6.1 + \log 20 = 6.1 + 1.3 = 7.4$$

⚡ The **ratio HCO_3^- : H_2CO_3 = 20:1** must be maintained for pH 7.4. Lungs control H_2CO_3 (via CO_2); kidneys control HCO_3^- . Change either $\rightarrow$ pH shifts.

BLOOD BUFFER SYSTEMS PYQ 2019, 2021

BUFFER SYSTEM	COMPONENTS	LOCATION	CAPACITY	KEY FEATURE
Bicarbonate/Carbonic acid	$\text{HCO}_3^- / \text{H}_2\text{CO}_3$ (= CO_2 dissolved)	Plasma + RBC	~53% of total	Open system — CO_2 expelled by lungs $\rightarrow$ most physiologically important. pKa 6.1
Haemoglobin	$\text{HbO}_2 / \text{HHb}$ (deoxyHb is better buffer)	RBC	~35% of total	Most powerful intracellular buffer. Histidine residues (pKa 6.0) — imidazole ring. DeoxyHb = stronger base than OxyHb $\rightarrow$ Haldane effect
Plasma proteins	Protein- NH_2 / Protein- COOH	Plasma	~7%	Amphoteric — can act as acid or base. Albumin main contributor
Phosphate	$\text{HPO}_4^{2-} / \text{H}_2\text{PO}_4^-$	RBC + urine	~5%	pKa 6.8 — ideal buffer for urine. Minor role in blood. Major role in renal H^+ excretion as titratable acid

🌸 MNEMONIC — BLOOD BUFFERS IN ORDER

"Big Healthy People Play" = Bicarbonate, Haemoglobin, Protein, Phosphate

🧬 HISTIDINE'S ROLE IN BUFFERING PYQ 2024

Histidine residues in haemoglobin have imidazole side chains with pKa ~6.0 — ideal for physiological buffering. The β -93 histidine is key. In deoxyHb, the pKa of histidine shifts to ~8.0 $\rightarrow$ stronger buffer $\rightarrow$ Bohr effect (H^+ promotes O_2 release).
Histidine is the only amino acid that buffers at physiological pH.

03

Acid-Base Disorders

4 PRIMARY ACID-BASE DISORDERS

★ PYQ MULTIPLE – 2013, 2014-S, 2018-S, 2023

DISORDER	PRIMARY CHANGE	PH	PCO ₂	HCO ₃ ⁻	COMPENSATION	COMMON CAUSES
Metabolic Acidosis	↓ HCO ₃ ⁻ (or ↑ H ⁺)	↓	Normal then ↓	↓ (primary)	Respiratory: ↑RR → ↓pCO ₂ (Kussmaul breathing)	DKA, lactic acidosis, renal failure, diarrhoea (HCO ₃ ⁻ loss), aspirin poisoning
Metabolic Alkalosis	↑ HCO ₃ ⁻	↑	Normal then ↑	↑ (primary)	Respiratory: ↓RR → ↑pCO ₂	Vomiting (HCl loss), antacid excess, hyperaldosteronism, diuretics, hypokalaemia
Respiratory Acidosis	↑ pCO ₂ (hypoventilation)	↓	↑ (primary)	Normal then ↑	Renal: ↑HCO ₃ ⁻ reabsorption + ↑H ⁺ excretion	COPD, respiratory depression (opioids), pneumonia, severe asthma, respiratory failure
Respiratory Alkalosis	↓ pCO ₂ (hyperventilation)	↑	↓ (primary)	Normal then ↓	Renal: ↓HCO ₃ ⁻ reabsorption + ↓H ⁺ excretion	Anxiety/hyperventilation, high altitude, salicylate poisoning (early), liver failure, sepsis

CLINICAL CASE – PERSISTENT VOMITING (ABG INTERPRETATION)

★ PYQ 2023



CASE: 50-YEAR-OLD MALE, PERSISTENT VOMITING, DEHYDRATED, SHALLOW BREATHING

Lab Results: pH 7.72 · HCO₃⁻ 45 mmol/L · pCO₂ 60 mmHg · Na⁺ 140 mEq/L · K⁺ 2.5 mEq/L · Urine: Acidic

► ABG INTERPRETATION APPROACH

Step 1: pH? pH 7.72 → **Alkalosis** (normal 7.35–7.45)

↓ Identify primary disturbance

Step 2: Primary cause? HCO_3^- 45 mEq/L (high) → Primary **Metabolic Alkalosis**
Cause: Vomiting → loss of HCl from stomach → HCO_3^- accumulates

↓ Check compensation

Step 3: Compensation? pCO_2 60 mmHg (expected ↑ to retain CO_2) — respiratory compensation: ↓RR → shallow breathing → CO_2 retention → helps bring pH back.
Appropriate respiratory compensation.

↓ Explain secondary findings

Step 4: Hypokalemia + Acidic urine — Paradoxical? **Why hypokalemia?** Vomiting → volume depletion → Aldosterone ↑ → Na^+ reabsorption + K^+ and H^+ excretion → K^+ loss
Why acidic urine in alkalosis? K^+ depleted cells exchange K^+ for H^+ → H^+ moves intracellularly + K^+ exits → remaining H^+ in distal tubule excreted in urine → paradoxically acidic urine despite systemic alkalosis (called "paradoxical aciduria")

● ABG QUICK RULE

"ROME" = Respiratory Opposite, Metabolic Equal

Respiratory acidosis → pH↓ and pCO_2 ↑ (opposite directions). Metabolic acidosis → pH↓ and HCO_3^- ↓ (same/equal direction). Applies for alkalosis too.

04 Anion Gap

ANION GAP — DEFINITION & SIGNIFICANCE

★ PYQ 2014-S, 2018-S

$$\text{Anion Gap} = [\text{Na}^+] - ([\text{Cl}^-] + [\text{HCO}_3^-])$$

Normal = 8–16 mEq/L (average 12 mEq/L)

Represents unmeasured anions: albumin (main), phosphate, sulphate, organic acids

TYPE	ANION GAP	MECHANISM	CAUSES
High Anion Gap Metabolic Acidosis (HAGMA)	↑ (>16 mEq/L)	Accumulation of unmeasured anions (organic acids, etc.) while HCO_3^- falls	MUDPILES: Methanol, Uraemia, DKA, Paraldehyde, Isoniazid/Iron, Lactic acidosis, Ethylene glycol, Salicylates
Normal Anion Gap Metabolic Acidosis (NAGMA / Hyperchloraemic acidosis)	Normal (8–16)	HCO_3^- lost (replaced by Cl^- to maintain electroneutrality) → Cl^- rises	Diarrhoea (GI HCO_3^- loss), RTA (Renal Tubular Acidosis), acetazolamide, ureterosigmoidostomy, saline infusion

● HAGMA MNEMONIC

"MUDPILES" = Methanol · Uraemia · DKA · Paraldehyde · Isoniazid/Iron · Lactic acidosis · Ethylene glycol · Salicylates

⚠ Hypoalbuminaemia reduces the anion gap by ~2.5 mEq/L for every 1 g/dL fall in albumin below 4 g/dL. Correct for albumin before interpreting anion gap!

05 Liver Function Tests (LFT)

⚡ Liver performs 500+ functions — LFTs assess hepatocellular damage, cholestasis, synthetic function, and excretory capacity. No single test is diagnostic alone — always interpret as a panel.

LFT PANEL — REFERENCE RANGES & CLINICAL SIGNIFICANCE

★ PYQ 2019, 2024

TEST	NORMAL RANGE	WHAT IT MEASURES	RAISED IN	PATTERN
Total Bilirubin	0.2–1.2 mg/dL	Total haem catabolism product	All types of jaundice	See jaundice section
Direct (Conjugated) Bilirubin	<0.2 mg/dL	Glucuronide-conjugated fraction	Obstructive, hepatocellular	↑↑ in obstructive/conjugated
Indirect (Unconjugated) Bilirubin	0.1–1.0 mg/dL	Albumin-bound fraction	Haemolytic jaundice, Gilbert's, neonatal	↑↑ in haemolytic/unconjugated
ALT / SGPT	7–56 U/L	Alanine aminotransferase — hepatocyte cytoplasm	Viral hepatitis (most sensitive), drug-induced liver injury	Most liver-specific. ALT > AST in viral hepatitis
AST / SGOT	10–40 U/L	Aspartate aminotransferase — liver + heart + muscle	Hepatitis, MI, muscle disease, alcoholic hepatitis	AST:ALT > 2 in alcoholic hepatitis
ALP (Alkaline Phosphatase)	44–147 U/L	Bile canaliculi enzyme, also bone/placenta	Obstructive jaundice (↑↑↑), bone disease, pregnancy	Best marker of cholestasis
GGT (γ-Glutamyl Transferase)	<55 U/L (M), <38 (F)	Bile duct cells, liver microsomes (induced by alcohol)	Obstructive jaundice, alcohol abuse, drug induction	High GGT + normal ALP = alcohol. High both = cholestasis
Total Protein	6.0–8.0 g/dL	Synthetic function	—	↓ in chronic liver disease (↓ synthesis)
Albumin	3.5–5.0 g/dL	Synthetic function, long half-life (~20 days)	—	↓ in chronic liver disease, malnutrition, nephrotic syndrome
Prothrombin Time (PT/INR)	11–14 sec (INR 0.8–1.2)	Clotting factors II, V, VII, X (liver-synthesised)	—	Prolonged in severe liver disease — sensitive marker of synthetic failure
5'-Nucleotidase	<17 U/L	Liver-specific cholestasis marker	Obstructive liver disease	Rises with ALP in cholestasis, not in bone disease — differentiates liver vs bone ALP

LFT PATTERN INTERPRETATION – PYQ 2024 CASE

 CASE: MIDDLE-AGED WOMAN, DARK URINE, GREY/PALE STOOL, PAIN ABDOMEN

Results: Total Bilirubin 6.2 mg/dL · Direct 5.8 mg/dL · Total Protein 7.2 g/dL · Albumin 3.8 g/dL · AST 26 U/L · ALT 32 U/L · ALP 387 U/L · GGT 35 U/L

Total Bilirubin 6.2 mg/dL ↑↑ Obstructive pattern: predominantly conjugated (direct)

Direct Bili 5.8 mg/dL ↑↑ Direct:Total ratio = 94% → obstructive/hepatocellular (not haemolytic)

ALT / AST Normal (32/26) No hepatocellular damage → not viral hepatitis; not alcoholic

ALP 387 U/L ↑↑↑ Markedly elevated → hallmark of cholestasis/biliary obstruction

Albumin 3.8 g/dL (normal) Synthetic function preserved → likely acute, not chronic

Clinical signs Dark urine + pale stool Classic obstructive jaundice: conjugated bili → urine (dark) + no bile to gut (pale stool)

Diagnosis: Obstructive (Cholestatic) Jaundice — likely choledocholithiasis (gallstone) or pancreatic head carcinoma. Further tests: USG abdomen, MRCP, ERCP, CA 19-9 if malignancy suspected.

WHY ALP RISES IN OBSTRUCTIVE JAUNDICE?

PYQ CLASSIC VIVA

► ALP MECHANISM IN CHOLESTASIS

Bile duct obstruction → bile salts accumulate in bile canaliculi

↓ Bile salts are detergents — solubilise membrane-bound enzymes

Bile salts solubilise ALP from bile canicular membranes → ALP released into blood

↓ Also

Bile salts induce ALP synthesis (transcriptional upregulation) in bile duct cells

↓ Result

Serum ALP rises markedly (3–10× normal) in obstructive jaundice

06 Jaundice & Bilirubin

⚡ Jaundice = yellowish discoloration of skin, sclerae, and mucous membranes when serum bilirubin >2 mg/dL. Normal serum bilirubin = 0.2–1.2 mg/dL. Clinical jaundice visible at >2 mg/dL.

► BILIRUBIN METABOLISM – COMPLETE PATHWAY

Step 1 — Haem Catabolism (Spleen/RES) Senescent RBCs → Haemoglobin → Haem + Globin

Haem → Haem oxygenase → **Biliverdin** (green) + CO + Fe²⁺
 Biliverdin → Biliverdin reductase → **Bilirubin** (yellow, water-insoluble = indirect/unconjugated)

↓ **Bound to albumin — transported in blood as indirect bilirubin**

Step 2 — Hepatic Uptake Indirect bilirubin (albumin-bound) → hepatocyte
 Dissociates from albumin → binds ligandin (Y protein) + Z protein inside hepatocyte

↓ **Conjugation in ER**

Step 3 — Conjugation Bilirubin + 2× UDP-glucuronic acid → **Bilirubin diglucuronide** (direct/conjugated bilirubin)

Enzyme: UDP-glucuronyl transferase (UGT1A1) — **absent in Crigler-Najjar; reduced in Gilbert's**

Conjugated bilirubin = water-soluble → can be excreted in bile

↓ **Active transport into bile canaliculi (MRP2 transporter) — rate-limiting step**

Step 4 — Intestinal Fate Bilirubin diglucuronide → intestinal bacteria → deconjugation → **Urobilinogen**

Urobilinogen: (a) 10–20% absorbed → portal blood → liver (enterohepatic circulation) or kidney → urine (gives urine yellow colour)

(b) Remainder → further reduced → **Stercobilin** → faeces (brown colour)

TYPES OF JAUNDICE – DIFFERENTIATION

★ PYQ 2019, 2024 – ALWAYS COMES

FEATURE	HAEMOLYTIC (PRE-HEPATIC)	HEPATOCELLULAR	OBSTRUCTIVE (POST-HEPATIC)
Cause	Excess RBC destruction	Hepatocyte damage (virus, toxin, alcohol)	Bile duct blockage (stone, tumour, stricture)
Total Bilirubin	↑	↑↑	↑↑↑
Indirect (Unconjugated)	↑↑ (mainly)	↑ both	Normal or slight ↑
Direct (Conjugated)	Normal	↑ both	↑↑↑ (mainly)
Urine Bilirubin	Absent (indirect can't cross kidney)	Present	Present (dark urine)
Urine Urobilinogen	↑↑↑ (excess Hb → excess urobilinogen)	↑ (impaired reuptake)	Absent (no bile reaching gut)
Stool Colour	Normal/dark	Normal or pale	Clay/pale (no stercobilin)
ALP	Normal	↑	↑↑↑
ALT/AST	Normal	↑↑↑ (hepatocellular damage)	Normal/slight ↑
Van den Bergh test	Indirect positive only	Both positive	Direct positive, delayed indirect

GILBERT'S SYNDROME

Mildly reduced UGT1A1 activity (~30% of normal) → mild unconjugated hyperbilirubinaemia, especially after fasting or stress. **Benign**, no liver disease. Serum bilirubin 1.2–3 mg/dL. Fasting test: ↑ bilirubin after 48-hour fast.

NEONATAL PHYSIOLOGICAL JAUNDICE + PHOTOTHERAPY

PYQ
2011,
2017-S

UGT1A1 immature at birth → unconjugated hyperbilirubinaemia → indirect bili (lipid-soluble) crosses BBB → kernicterus. **Phototherapy**: blue light (420–480 nm) converts insoluble bilirubin → lumirubin/photobilirubin (water-soluble, polar isomers) → excreted without conjugation in bile and urine.

07 Renal Function Tests (RFT)

GLOMERULAR FILTRATION TESTS

★ PYQ 2012-S, 2014, 2018, 2022, 2024

TEST	NORMAL VALUE	CLINICAL SIGNIFICANCE	KEY POINTS
Serum Creatinine	0.6–1.2 mg/dL (M) 0.5–1.0 (F)	GFR estimate. More specific than urea for renal function	Produced at constant rate from muscle creatine. NOT affected by diet or protein intake (unlike urea). Rises late — 50% nephron loss before serum Cr rises
Blood Urea Nitrogen (BUN)	7–20 mg/dL (urea) 15–40 mg/dL	Glomerular function, protein catabolism	Affected by protein intake, dehydration, GI bleeding (pre-renal). BUN:Cr ratio >20 = pre-renal; <10 = malnutrition
Creatinine Clearance (CrCl)	85–125 mL/min (M) 75–115 (F)	Best clinical estimate of GFR	Formula: $CrCl = (\text{Urine Cr} \times \text{Volume}) / \text{Serum Cr}$. MDRD and CKD-EPI formulas also used. 24-hour urine collection
eGFR	>90 mL/min/1.73m ²	CKD staging (Stage 1–5)	CKD Stage 3: eGFR 30–59; Stage 4: 15–29; Stage 5 (ESRD): <15
Uric Acid	3.5–7.2 mg/dL (M) 2.6–6.0 (F)	Gout, Von Gierke's, CKD	Elevated in renal failure, gout, high cell turnover (leukaemia)

WHY CREATININE IS MORE SPECIFIC THAN UREA

★ PYQ 2022

Creatinine:

Urea (BUN) is affected by:

- ◆ Produced at constant rate from creatine phosphate in muscle
- ◆ Freely filtered at glomerulus
- ◆ Minimal tubular secretion (slightly overestimates GFR)
- ◆ NOT affected by protein diet, hydration, or GI bleeding
- ◆ More specific for renal function
- ◆ High-protein diet → ↑ urea (pre-renal false positive)
- ◆ GI bleeding → protein absorbed → ↑ urea
- ◆ Dehydration → ↑ urea
- ◆ Liver failure → ↓ urea (decreased synthesis)
- ◆ Tubular reabsorption of urea increases with dehydration

🔗 UREMIA & AMMONIA TOXICITY – PYQ 2024

Chronic renal failure → urea accumulates → gut bacteria convert urea → NH_3 → absorbed → exceeds liver detoxification → hyperammonaemia → inhibits α -KG, glutamate, ATP in brain → encephalopathy (asterixis, confusion). Also: metabolic acidosis (cannot excrete H^+ + loss of HCO_3^- reabsorption), anaemia (↓ erythropoietin), hyperphosphataemia + hypocalcaemia (↓ calcitriol), uraemic frost.

TUBULAR FUNCTION TESTS

TEST	NORMAL	RAISED/ABNORMAL IN
Urine concentration test (osmolality)	800–1200 mOsm/kg after dehydration	Tubular damage: cannot concentrate → dilute urine
Urine sodium	<20 mEq/L (pre-renal), >40 (renal)	Differentiates pre-renal from intrinsic renal failure
FENa (Fractional excretion of Na^+)	<1% = pre-renal; >2% = ATN	Most useful in distinguishing pre-renal vs ATN
β_2-Microglobulin (urine)	<0.3 mg/L	Tubular damage marker (proximal tubule fails to reabsorb)
Renal clearance (general)	CrCl as above	PYQ 2014, 2018 – "Renal clearance is predictor of impending renal failure"

08

Diabetes Lab Tests

DIAGNOSTIC CRITERIA (WHO/ADA) & TESTS

★ PYQ 2011-S, 2016, 2018

TEST	NORMAL	PRE-DIABETES	DIABETES MELLITUS
Fasting plasma glucose (FPG)	<100 mg/dL	100–125 (IFG)	≥126 mg/dL (on 2 occasions)
2-hr post-OGTT (75 g glucose)	<140 mg/dL	140–199 (IGT)	≥200 mg/dL
Random plasma glucose	<140 mg/dL	—	≥200 mg/dL + symptoms
HbA1c	<5.7%	5.7–6.4%	≥6.5%

HbA1c – GLYCATED HAEMOGLOBIN

★ PYQ 2011, 2016, 2018

Mechanism:

- ◆ Glucose attaches non-enzymatically to N-terminal valine of β -globin chains (Hb) → **glycation** (Amadori product)
- ◆ Rate of glycation is proportional to ambient blood glucose over the lifespan of the RBC
- ◆ RBC lifespan = 120 days → HbA1c reflects **average blood glucose over 2–3 months**
- ◆ HbA1c rises 0.3% for every 30 mg/dL rise in mean plasma glucose

Clinical significance:

- ◆ HbA1c <7% = well-controlled diabetes (target)
- ◆ NOT affected by short-term glucose fluctuations
- ◆ Falsely LOW in haemolytic anaemia (short RBC lifespan), haemorrhage, HbS trait
- ◆ Falsely HIGH in iron deficiency anaemia (long RBC lifespan), splenectomy
- ◆ Not useful in haemoglobinopathies (HbS, HbC) — use fructosamine instead

ORAL GLUCOSE TOLERANCE TEST (OGTT)

PYQ 2011-S

► OGTT PROTOCOL

Preparation 3 days of high-carbohydrate diet (≥150 g/day) + overnight fast (8–14 hours) + no exercise, smoking, drugs

↓ 0 min: Fasting sample taken

Load 75 g anhydrous glucose in 250 mL water (adult) · 1.75 g/kg in children (max 75 g) · Drink within 5 minutes

↓ Blood samples at 30, 60, 90, 120 minutes

Interpretation Normal: 2-hr value <140 mg/dL · IGT (pre-DM): 140–199 · DM: ≥200 mg/dL

09 Free Radicals & ROS

⚡ A free radical = any species with one or more unpaired electrons in its outermost orbital → highly reactive, short-lived, chain-reaction propagating. Reactive Oxygen Species (ROS) = oxygen-centred free radicals + reactive non-radical oxygen derivatives.

MAJOR ROS – PROPERTIES & SOURCES

★ PYQ 2013, 2015-S, 2019

ROS	SYMBOL	SOURCE	PROPERTIES
Superoxide radical	$O_2^{\bullet -}$	ETC (Complex I, III leak), NADPH oxidase (phagocytes), xanthine oxidase	Moderately reactive. Cannot cross membranes easily. Precursor to H_2O_2 and $OH^{\bullet}$
Hydrogen Peroxide	H_2O_2	SOD reaction, peroxisomes (fatty acid oxidation), monoamine oxidase	Not a radical (no unpaired electron) but ROS. Can cross membranes. Generates $OH^{\bullet}$ via Fenton reaction
Hydroxyl radical	$OH^{\bullet}$	Fenton reaction: $Fe^{2+} + H_2O_2 \rightarrow Fe^{3+} + OH^{\bullet} + OH^-$ (Haber-Weiss)	Most reactive and most damaging ROS. No enzyme scavenger — only quenched by vitamin C, E, or contained by antioxidants
Singlet oxygen	1O_2	Photosensitisation reactions, UV light	Excited state of O_2 . Reacts with unsaturated fatty acids
Nitric oxide	$NO^{\bullet}$	Nitric oxide synthase (NOS) from arginine	Free radical — also vasodilator. Can form peroxynitrite ($ONOO^-$) with $O_2^{\bullet -} \rightarrow$ cell damage
Hypochlorous acid	$HOCl$	Myeloperoxidase in neutrophils ($H_2O_2 + Cl^-$)	Potent bactericidal agent in phagocytes

► FREE RADICAL CHAIN REACTION – LIPID PEROXIDATION

Initiation $OH^{\bullet} + \text{Polyunsaturated Fatty Acid (LH)} \rightarrow L^{\bullet} \text{ (lipid radical)} + H_2O$

↓ Propagation — chain reaction

Propagation $L^{\bullet} + O_2 \rightarrow LOO^{\bullet}$ (lipid peroxy radical)
 $LOO^{\bullet} + LH \text{ (new fatty acid)} \rightarrow LOOH \text{ (lipid hydroperoxide)} + L^{\bullet}$
 Chain reaction continues — ONE radical can destroy many fatty acids

↓ Termination by antioxidants (Vit E)

Termination $LOO^{\bullet} + \text{Vitamin E} \rightarrow LOOH + \text{Vit E}^{\bullet}$ (tocopheroxyl radical, less reactive)
 $\text{Vit E}^{\bullet} + \text{Vit C} \rightarrow \text{Vit E regenerated} + \text{ascorbate radical}$
 Chain reaction stops. TBARS (malondialdehyde) formed — used as marker of lipid peroxidation

ROS – BENEFICIAL EFFECTS

PYQ 2015-S, 2021

⚡ **Respiratory burst (oxidative burst):** NADPH oxidase in neutrophils/macrophages → $O_2^{\bullet -} \rightarrow HOCl$ (myeloperoxidase) → kills bacteria. CGD (Chronic Granulomatous Disease) = NADPH

oxidase defect → recurrent infections

♦ **Cell signalling:** H_2O_2 acts as second messenger in insulin signalling, NF- κ B activation, growth factor receptor signalling

♦ **Thyroid hormone synthesis:** H_2O_2 generated by DUOX (thyroid oxidase) is required for iodination of thyroglobulin

♦ **Nitric oxide:** $\text{O}_2^{\bullet-}$ -independent — vasodilation, neurotransmission, immune function

♦ **Sperm capacitation:** ROS generated by sperm NADPH oxidase required for hyperactivation

10 Antioxidants & Detoxification

ENZYMATIC ANTIOXIDANTS

★ PYQ 2013, 2014, 2019

ENZYME	REACTION	LOCATION	COFACTOR	SIGNIFICANCE
Superoxide Dismutase (SOD)	$2\text{O}_2^{\bullet-} + 2\text{H}^+ \rightarrow \text{H}_2\text{O}_2 + \text{O}_2$	Cytoplasm (Cu/Zn-SOD), Mitochondria (Mn-SOD), Extracellular (Cu/Zn-SOD)	Cu, Zn (cytoplasmic); Mn (mitochondrial)	First line of enzymatic defence. SOD1 mutation → familial ALS. SOD converts $\text{O}_2^{\bullet-}$ (reactive) → H_2O_2 (less reactive but still dangerous)
Catalase	$2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$	Peroxisomes mainly, RBCs, liver	Iron (haem-containing)	Detoxifies H_2O_2 in peroxisomes. Converts H_2O_2 generated during fatty acid β -oxidation. Important in RBCs (no mitochondria)
Glutathione Peroxidase (GPx)	$\text{ROOH} + 2\text{GSH} \rightarrow \text{ROH} + \text{GSSG} + \text{H}_2\text{O}$ $\text{H}_2\text{O}_2 + 2\text{GSH} \rightarrow 2\text{H}_2\text{O} + \text{GSSG}$	Cytoplasm + mitochondria	Selenium (Se) — essential trace mineral	Most important H_2O_2 scavenger in cytoplasm. Requires GSH (reduced glutathione). Se deficiency → ↓ GPx → Keshan disease (cardiomyopathy)
Glutathione Reductase	$\text{GSSG} + \text{NADPH} \rightarrow 2\text{GSH}$	Cytoplasm	NADPH (from HMP shunt)	Regenerates reduced GSH. Links HMP shunt to ROS defence. G6PD deficiency → ↓ NADPH → ↓ GSH → ↑ oxidative haemolysis
Thioredoxin Reductase	$\text{Trx(ox)} + \text{NADPH} \rightarrow \text{Trx(red)}$	Cytoplasm, mitochondria	Se, FAD, NADPH	Backup to glutathione system

NON-ENZYMATIC (DIETARY) ANTIOXIDANTS

★ PYQ 2011-S, 2012-S, 2022

ANTIOXIDANT	MECHANISM	SOURCE	DEFICIENCY / CLINICAL NOTE
Vitamin E (α-Tocopherol)	Chain-breaking antioxidant — donates $H\cdot$ to $LOO\cdot$ → terminates lipid peroxidation chain. Lipid-soluble → in membranes	Vegetable oils, nuts, wheat germ	Deficiency → haemolytic anaemia (in premature neonates), ataxia, neuropathy. Regenerated by Vit C
Vitamin C (Ascorbic acid)	Water-soluble radical scavenger. Donates electrons to Vit E radical → regenerates Vit E. Scavenges $O_2\cdot^-$, $OH\cdot$, H_2O_2 . Required for prolyl hydroxylase (collagen)	Citrus fruits, vegetables	Deficiency → Scurvy. Synergistic with Vit E
β-Carotene (pro-Vit A)	Quenches singlet oxygen (1O_2). Singlet oxygen trapper. Antioxidant especially under low pO_2	Orange/yellow vegetables, carrots	High dose supplements may ↑ lung cancer risk in smokers (pro-oxidant at high pO_2). Converted to retinal in gut
Glutathione (GSH)	Tripeptide (Gly-Cys-Glu). Thiol ($-SH$) donates electrons. Substrate for GPx. Conjugates with electrophiles (Phase II detox)	Synthesised endogenously (liver)	N-acetylcysteine (NAC) replenishes GSH in paracetamol overdose. Key in detoxification and oxidative defence
Ubiquinol (CoQ-H₂)	Reduced form of CoQ10 — scavenges lipid radicals in inner mitochondrial membrane	Endogenous (mevalonate pathway)	Statins ↓ CoQ10 synthesis (block mevalonate) → myopathy in some patients
Selenium	Integral part of GPx and thioredoxin reductase (selenocysteine residue)	Meat, seafood, nuts	Keshan disease (cardiomyopathy — Se-deficient soil in China)
Uric acid	Scavenges $OH\cdot$, singlet oxygen, peroxynitrite in plasma	Purine catabolism	Major aqueous-phase antioxidant in human plasma (we lost uricase evolutionarily — may confer antioxidant advantage)
Ceruloplasmin	Ferroxidase activity — oxidises Fe^{2+} → Fe^{3+} → prevents Fenton reaction	Plasma glycoprotein	Wilson's disease (↓ ceruloplasmin) → ↑ free Cu^{2+} → ↑ oxidative damage

▶ GLUTATHIONE CYCLE — ANTIOXIDANT-DETOX HUB

HMP Shunt → NADPH G6PD: $G6P \rightarrow 6-PG + NADPH$ (essential — only source in RBC)

↓ NADPH powers glutathione reductase

Glutathione Reductase $GSSG$ (oxidised) + NADPH → 2 GSH (reduced) — maintains pool of reduced glutathione

↓ GSH used by GPx to destroy peroxides

Glutathione Peroxidase (GPx) $H_2O_2 + 2GSH \rightarrow 2H_2O + GSSG$
 $ROOH + 2GSH \rightarrow ROH + GSSG + H_2O$ (lipid peroxides neutralised)

↓ Also — Phase II detox conjugation

Glutathione S-Transferase (GST) $GSH + \text{electrophilic toxin} \rightarrow GSH \text{ conjugate} \rightarrow$
 mercapturic acid → excreted in urine
 Detoxification of paracetamol, bilirubin, carcinogens

🔗 PARACETAMOL (ACETAMINOPHEN) OVERDOSE — GSH CONNECTION

PYQ CLASSIC

Therapeutic dose: paracetamol → glucuronide/sulphate conjugates (Phase II).
 Overdose → CYP2E1 (Phase I) → NAPQI (toxic reactive metabolite). Normal: $NAPQI + GSH \rightarrow$ non-toxic mercapturic acid. Overdose: GSH depleted → NAPQI binds hepatocyte proteins → hepatocellular necrosis → fulminant hepatic failure. **Treatment: N-Acetylcysteine (NAC)** — replenishes GSH by providing cysteine → cures paracetamol hepatotoxicity if given early.

11 Xenobiotics & Cytochrome P450

- ⚡ Xenobiotic = any chemical foreign to the body (drugs, pollutants, food additives, carcinogens).
 Biotransformation = enzymatic conversion of xenobiotics to more polar, excretable metabolites. Occurs mainly in liver (SER). Also in intestine, lung, kidney, skin.

PHASES OF XENOBIOTIC METABOLISM

★ PYQ 2016-S, 2023, 2025

► BIOTRANSFORMATION – 2 PHASES (PHASE III SOMETIMES LISTED)

PHASE I — Functionalisation (reveal or add reactive group) Reactions: Oxidation (most common), Reduction, Hydrolysis

Enzymes: **Cytochrome P450 (CYP)** monooxygenases (main), monoamine oxidase, alcohol dehydrogenase, esterases

Products: More polar, often still toxic. **NAPQI from paracetamol** = classic toxic Phase I product

Location: Smooth ER (microsomes) of hepatocytes

Makes drug more reactive — NOT always detoxification!

↓ Phase I product → Phase II conjugation

PHASE II — Conjugation (attach a polar group) Reactions: **Glucuronidation** (most common, uses UDPGA) · Sulphation · Acetylation · Methylation · Glutathione conjugation · Glycine conjugation

Products: Highly polar, non-toxic, excretable in urine or bile

Location: Cytosol + ER (glucuronidation in ER)

Usually detoxification. Some conjugates still active (morphine-6-glucuronide is more potent than morphine)

↓ (Optional Phase III — transport)

PHASE III — Elimination/Transport MDR transporters (P-glycoprotein), MRP2, OATP → excrete conjugates into bile or urine

Important for multi-drug resistance in cancer

CYTOCHROME P450 SYSTEM

★ PYQ 2013, 2013-S, 2014-S, 2019, 2025

Structure:

- ◆ Haemoprotein (iron-containing haem) — absorbs light at 450 nm when CO-bound (hence P450)
- ◆ Located in smooth ER (microsomes)
- ◆ Requires NADPH + O₂ (monooxygenase reaction)
- ◆ Reaction: $\text{RH} + \text{O}_2 + \text{NADPH} + \text{H}^+ \rightarrow \text{ROH} + \text{H}_2\text{O} + \text{NADP}^+$
- ◆ Electron donors: NADPH-CYP reductase (FAD, FMN) + cytochrome b₅

Important CYP isoforms:

- ◆ **CYP1A2**: Caffeine, theophylline, warfarin
- ◆ **CYP2D6**: Codeine, β-blockers, antidepressants (polymorphic — poor/extensive metabolisers)
- ◆ **CYP2E1**: Ethanol, paracetamol (generates NAPQI), isoniazid
- ◆ **CYP3A4**: Most abundant CYP (~50% of drug metabolism) — statins, cyclosporin, midazolam

CYP Inducers (↑ CYP activity → ↓ drug effect):

- ◆ Rifampicin, Phenobarbitone, Phenytoin, Carbamazepine, Alcohol (chronic), St John's Wort
- ◆ ↑ CYP → ↑ drug metabolism → ↓ therapeutic effect (enzyme induction)
- ◆ Also → ↑ sex steroid metabolism (OCP failure with rifampicin)

CYP Inhibitors (↓ CYP activity → ↑ drug effect/toxicity):

- ◆ Ketoconazole, Erythromycin, Cimetidine, Grapefruit juice (CYP3A4), Isoniazid, Fluconazole, Acute alcohol
- ◆ ↓ CYP → ↓ drug metabolism → ↑ plasma levels → toxicity

PHASE II REACTIONS – CONJUGATES

PYQ 2019-S, 2023

REACTION	CONJUGATING AGENT	ENZYME	EXAMPLE SUBSTRATE	PRODUCT EXCRETED VIA
Glucuronidation MOST COMMON	UDP-glucuronic acid (UDPGA)	UDP-glucuronyl transferase (UGT)	Bilirubin, morphine, paracetamol, steroids, chloramphenicol	Bile + urine
Sulphation	PAPS (3'-phosphoadenosine-5'-phosphosulphate)	Sulphotransferase	Paracetamol (therapeutic), steroids	Urine
Glutathione conjugation	Reduced glutathione (GSH)	Glutathione S-transferase	NAPQI, halogenated hydrocarbons, epoxides	Mercapturic acid → urine
Acetylation	Acetyl-CoA	N-acetyltransferase (NAT) — polymorphic	Isoniazid, sulphonamides, dapsone	Urine. Slow/fast acetylators
Glycine conjugation	Glycine + CoA	Glycine N-acyltransferase	Benzoic acid → hippuric acid, bile acids	Urine
Methylation	SAM (S-adenosylmethionine)	Methyltransferases	Catecholamines (COMT), histamine (HNMT)	Urine

● PHASE II REACTIONS MNEMONIC

"GluSAGMe" = Glucuronidation · Sulphation · Acetylation · Glutathione · Methylation

12 ELISA & Functional Tests

ELISA — ENZYME-LINKED IMMUNOSORBENT ASSAY

★ PYQ 2014-S

► SANDWICH ELISA — MOST COMMON TYPE

Step 1 — Coat Capture antibody (Ab₁) coated on microtitre plate wells

↓ Block with BSA to prevent non-specific binding

Step 2 — Add sample Patient sample added → antigen binds to capture Ab₁ → wash away unbound

↓

Step 3 — Detection antibody Enzyme-linked detection Ab₂ added → binds antigen (different epitope) → wash away unbound

↓

Step 4 — Substrate & Read Substrate added → enzyme (HRP or ALP) converts → coloured product
OD measured at 450 nm → proportional to antigen concentration

ELISA types:

- ◆ **Sandwich ELISA:** Quantifies antigen between 2 antibodies (most common)
- ◆ **Indirect ELISA:** Detects antibodies in patient serum (HIV screening — ELISA for anti-HIV antibodies)
- ◆ **Competitive ELISA:** Antigen in sample competes with labelled antigen
- ◆ **Capture ELISA:** Variant of sandwich, for low-abundance antigens

Applications:

- ◆ HIV screening (anti-HIV antibodies)
- ◆ HBsAg (Hepatitis B surface antigen)
- ◆ Troponin (AMI diagnosis)
- ◆ Pregnancy test (β-hCG)
- ◆ Tumour markers (PSA, AFP, CA 125)
- ◆ Hormones (TSH, LH, FSH)
- ◆ Drug levels (digoxin, cyclosporin)

CHROMATOGRAPHY PRINCIPLES

PYQ 2017, 2016-S, 2019-S

TYPE	STATIONARY PHASE	MOBILE PHASE	PRINCIPLE	APPLICATION
Ion Exchange Chromatography	Charged resin (cation: Sephadex; anion: DEAE)	Buffer of varying ionic strength	Molecules separated by charge. Cation exchangers bind +ve molecules; anion exchangers bind -ve	Amino acid analysis, protein purification, HbA1c (Hb variants)
Gel Filtration (Size Exclusion)	Porous beads (Sephadex, Sepharose)	Buffer	Small molecules enter pores (slow), large excluded (fast). Separates by molecular size/weight	Molecular weight determination, desalting
Affinity Chromatography	Ligand covalently linked to matrix (e.g., lectin, antibody, substrate analogue)	Buffer → elute with competing ligand	Highly specific binding between molecule and immobilised ligand. Most specific method	Antibody purification, enzyme isolation, HbA1c (lectin affinity)
Thin Layer Chromatography (TLC)	Silica gel on glass/aluminium plate	Organic solvent (mobile)	R_f = distance of spot / distance of solvent front. Separates by polarity	Lipid separation (phospholipids), drug screening, vitamins
HPLC	Fine silica particles	Pressurised solvent	High resolution separation at high pressure; UV/fluorescence detection	Drug analysis, HbA1c (gold standard)

13 Viva Q&A

ACID-BASE BALANCE

Write the Henderson-Hasselbalch equation. How does the kidney maintain acid-base balance?PYQ 2019-S

$$\text{pH} = \text{pK}_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

For bicarbonate buffer: $\text{pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$

Normal ratio = 20:1 → pH 7.4. Kidneys maintain this by:

- **HCO₃⁻ reabsorption:** Tubular cells secrete H⁺ into lumen → H⁺ + filtered HCO₃⁻ → H₂CO₃ → CO₂ → enters cell → recombines → HCO₃⁻ exits basolateral side into blood. Catalysed by carbonic anhydrase at both steps
- **Titratable acid excretion:** H⁺ + HPO₄²⁻ → H₂PO₄⁻ in tubular lumen → excreted. Contributes ~20–30 mEq H⁺/day
- **NH₄⁺ excretion:** Glutamine → glutaminase → NH₃ + H⁺ → NH₄⁺ → excreted. Increases dramatically (10×) in chronic acidosis — most adaptable system

Exam tip: Always state that only the kidney can COMPLETELY restore pH (not just buffer). Respiratory compensation is faster but incomplete.

What is anion gap? What is its biomedical importance?PYQ 2014-S, 2018-S

$$\text{Anion gap} = [\text{Na}^+] - ([\text{Cl}^-] + [\text{HCO}_3^-]) = 8\text{--}16 \text{ mEq/L}$$

Represents unmeasured anions (albumin, phosphate, sulphate, organic acids) that maintain electroneutrality.

Importance:

- Differentiates causes of metabolic acidosis: High AG = accumulation of unmeasured anions (MUDPILES); Normal AG = HCO₃⁻ loss with Cl⁻ replacement (diarrhoea, RTA)
- Guides treatment: High AG (DKA) → insulin + fluids; High AG (aspirin) → alkalinisation of urine
- Prognosis: Very high AG (>30) suggests severe lactic acidosis → poor prognosis

Correct for albumin: every 1 g/dL drop in albumin below 4 → reduce normal AG by 2.5 mEq/L

 LIVER & RENAL FUNCTION

Why does ALP increase in obstructive jaundice but not haemolytic jaundice? PYQ Classic Viva

ALP in obstructive jaundice:

- ALP is normally membrane-bound in bile canaliculi of hepatocytes
- Obstruction → bile salts accumulate in canaliculi → bile salts (detergents) solubilise membrane lipids → release membrane-bound ALP into blood
- Also: bile salts induce increased ALP synthesis (transcription) in bile duct epithelium
- Result: ALP rises markedly (3–10× normal)

Why NOT in haemolytic jaundice: No bile duct involvement, no bile salt accumulation → ALP not released. Liver's synthetic and excretory function intact.

Key: 5'-nucleotidase distinguishes liver ALP from bone ALP — rises only with liver origin

Why is serum creatinine more specific than plasma urea for renal function? PYQ 2022

Creatinine superiority:

- Produced at constant rate from creatine phosphate in muscle (not affected by diet)
- Freely filtered at glomerulus, minimal tubular secretion → accurate GFR estimate
- Not reabsorbed by tubules

Urea limitations:

- ↑ by: high protein diet, GI bleeding (protein absorbed), dehydration (↑ tubular reabsorption), catabolic states
- ↓ by: liver failure (↓ synthesis), malnutrition, pregnancy
- 20–40% reabsorbed in proximal tubule (passively) → influenced by flow rate

However: serum Cr rises LATE — 50% of nephron mass lost before Cr rises above normal. Creatinine clearance more sensitive early marker.

FREE RADICALS, ANTIOXIDANTS & XENOBIOTICS

What is the role of Superoxide Dismutase? How does it protect cells? PYQ 2013

SOD reaction: $2\text{O}_2^{\bullet-} + 2\text{H}^+ \rightarrow \text{H}_2\text{O}_2 + \text{O}_2$ (dismutation)

Protection mechanism:

- $\text{O}_2^{\bullet-}$ itself moderately reactive but becomes dangerous by generating $\text{OH}^\bullet$ via Haber-Weiss/Fenton reactions ($\text{Fe}^{2+} + \text{H}_2\text{O}_2 \rightarrow \text{Fe}^{3+} + \text{OH}^\bullet$)
- SOD converts $\text{O}_2^{\bullet-}$ to H_2O_2 (less reactive, cannot generate $\text{OH}^\bullet$ without Fe^{2+})
- H_2O_2 then detoxified by catalase (peroxisomes) or GPx (cytoplasm)
- Cu/Zn-SOD = cytoplasm; Mn-SOD = mitochondria (where most $\text{O}_2^{\bullet-}$ is generated)

Clinical link:

- SOD1 (Cu/Zn-SOD) mutations → Familial ALS (amyotrophic lateral sclerosis) — motor neuron disease
- SOD3 deficiency → ↑ cardiovascular oxidative damage

SOD is the first line of enzymatic antioxidant defence — without it, $\text{O}_2^{\bullet-}$ accumulates → reacts with $\text{NO}^\bullet$ → ONOO^- (peroxynitrite) → cell death

Explain Phase I and Phase II reactions of xenobiotic metabolism with examples. PYQ 2016-S, 2023

Phase I (Functionalisation):

- Goal: Introduce or unmask a reactive functional group ($-\text{OH}$, $-\text{NH}_2$, $-\text{SH}$) → makes molecule more polar and reactive for Phase II
- Main reaction: Oxidation by CYP450 monooxygenases (SER). Also: reduction, hydrolysis
- NOT always detoxification — can activate procarcinogens (benzo[a]pyrene → epoxide → DNA damage)
- Example: Paracetamol → CYP2E1 → NAPQI (toxic); Codeine → CYP2D6 → Morphine (active)

Phase II (Conjugation):

- Goal: Conjugate with large polar molecule → high MW, very water-soluble → excrete in bile or urine
- Reactions: Glucuronidation (most common), sulphation, acetylation, glutathione conjugation, glycine conjugation, methylation
- Generally detoxifying — produces non-toxic, excretable products
- Example: Morphine + UDPGA → morphine-6-glucuronide (paradoxically MORE potent); Bilirubin → bilirubin diglucuronide (water-soluble → bile)

Exam trap: Phase I is NOT always detoxification. Phase II is generally (but not always) detoxification.

What is glutathione? Describe its role in detoxification. PYQ 2014-S, 2016-S, 2022

Glutathione (GSH): Tripeptide — γ -Glutamyl-Cysteinyl-Glycine. The γ -bond (unusual) protects from non-specific proteolysis.

Roles in detoxification:

- **Direct radical scavenging:** $\text{GSH} + \text{OH}^\bullet \rightarrow \text{GS}^\bullet + \text{H}_2\text{O} \rightarrow \text{GS}^\bullet + \text{GS}^\bullet \rightarrow \text{GSSG}$ (oxidised glutathione)
- **GPx substrate:** $\text{H}_2\text{O}_2 + 2\text{GSH} \rightarrow 2\text{H}_2\text{O} + \text{GSSG}$; $\text{ROOH} + 2\text{GSH} \rightarrow \text{ROH} + \text{GSSG} + \text{H}_2\text{O}$
- **Phase II conjugation:** $\text{GSH} + \text{electrophilic toxin (NAPQI, epoxides)} \rightarrow \text{glutathione conjugate} \rightarrow \text{cysteinyl conjugate} \rightarrow \text{mercapturic acid} \rightarrow \text{excreted}$. Protects DNA and proteins from reactive metabolites
- **RBC protection:** $\downarrow \text{GSH (G6PD deficiency)} \rightarrow \downarrow \text{NADPH} \rightarrow \downarrow \text{GSH regeneration} \rightarrow \text{Heinz body formation} \rightarrow \text{haemolytic anaemia}$
- **Leukotriene synthesis:** $\text{GSH} + \text{arachidonic acid pathway} \rightarrow \text{leukotrienes (LTC}_4, \text{ LTD}_4, \text{ LTE}_4) \rightarrow \text{inflammatory mediators}$

N-Acetylcysteine (NAC) is the antidote for paracetamol overdose because it donates cysteine $\rightarrow$ replenishes GSH $\rightarrow$ binds NAPQI before it destroys hepatocytes

 DIABETES & JAUNDICE**Why is phototherapy used to cure neonatal/physiological jaundice?PYQ 2011, 2017-S**

Background: Neonates have immature UGT1A1 → cannot conjugate bilirubin adequately → unconjugated (indirect) bilirubin accumulates. At high levels → crosses BBB (lipid-soluble) → deposits in basal ganglia → kernicterus (irreversible brain damage).

Why phototherapy works:

- Blue-green light (420–490 nm; peak efficacy 460 nm) photoexcites unconjugated bilirubin in skin capillaries
- Two photochemical reactions: (a) Photoisomerisation → 4Z,15E-bilirubin (configurational isomer) → more polar, excreted in bile WITHOUT conjugation; (b) Photooxidation → lumirubin (structural isomer) → water-soluble, excreted in urine and bile
- Both products water-soluble → can be excreted without conjugation → safe (cannot cross BBB)

Key point: No liver function required for phototherapy efficacy — therefore works even in UGT-deficient neonates (Crigler-Najjar Type II). Blue light specifically (not white) — hence blue lights in neonatal wards.

How does HbA1c provide information for management of diabetes mellitus?PYQ 2016

Mechanism: Glucose non-enzymatically glycates N-terminal valine of β -globin → HbA1c (Amadori rearrangement). Rate $\propto$ ambient blood glucose $\times$ time.

Why it's useful:

- RBC lifespan = 120 days → HbA1c reflects **mean blood glucose over 2–3 months** → not affected by today's meal, stress, or glucose fluctuation
- Fasting blood glucose reflects only the past 8 hours; HbA1c gives the long-term picture
- Target HbA1c $<7\%$ → associated with $\downarrow$ microvascular complications (DCCT, UKPDS trials)
- Every 1% rise in HbA1c → $\uparrow$ risk of MI by 14%, retinopathy by 37%

Limitations: Falsely low in haemolytic anaemia, sickle cell disease (short RBC lifespan). Falsely high in iron deficiency anaemia. Not reliable in haemoglobinopathies → use fructosamine instead.

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Essay Question Templates

EQ — ACID-BASE BALANCE & ANION GAP [PYQ 2014-S, 2018-S]

"Describe how pH of blood is regulated by lung and kidney. Give an account of metabolic acidosis and anion gap."

- 1. Normal blood pH:** 7.35–7.45. Maintained by three systems acting in sequence — buffers (seconds), respiratory (minutes), renal (hours–days).
- 2. Respiratory regulation:** Carotid/medullary chemoreceptors sense $\uparrow \text{H}^+$ (or $\uparrow \text{CO}_2$) $\rightarrow$ stimulate respiratory centre $\rightarrow \uparrow \text{RR} \rightarrow$ more CO_2 expelled $\rightarrow [\text{H}_2\text{CO}_3]$ falls $\rightarrow$ pH rises. The converse in alkalosis. Lungs maintain the CO_2 side of the Henderson-Hasselbalch equation.
- 3. Renal regulation:** Kidneys maintain HCO_3^- concentration. Mechanisms: (i) HCO_3^- reabsorption (proximal tubule, carbonic anhydrase-dependent); (ii) Titratable acid excretion (H^+ + phosphate $\rightarrow \text{H}_2\text{PO}_4^-$); (iii) Ammonium excretion (glutamine $\rightarrow \text{NH}_3 \rightarrow \text{NH}_4^+$) — increases 10-fold in chronic acidosis. Only mechanism for COMPLETE pH correction.
- 4. Metabolic Acidosis:** Primary $\downarrow \text{HCO}_3^- \rightarrow$ pH falls $\rightarrow$ respiratory compensation (Kussmaul breathing, $\downarrow \text{pCO}_2$). Causes: DKA, lactic acidosis, renal failure, diarrhoea, salicylate poisoning. HbA1c provides long-term glucose monitoring.
- 5. Anion Gap:** $\text{AG} = \text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-) = 8\text{--}16 \text{ mEq/L}$. Represents unmeasured anions. High AG metabolic acidosis (MUDPILES) = accumulation of organic acids displacing HCO_3^- . Normal AG acidosis = HCO_3^- loss with compensatory Cl^- retention (hyperchloraemic).
- 6. Clinical significance of AG:** Distinguishes cause of metabolic acidosis without additional tests. Guides treatment — DKA (insulin); lactic acidosis (treat sepsis/hypoxia); methanol (fomepizole, dialysis). Very high AG (>30) = severe $\rightarrow$ poor prognosis.

 EQ – FREE RADICALS HAVE BENEFICIAL ROLE [PYQ 2015-S, 2021]

"Free radicals may have some beneficial role also."

1. **Definition:** Free radicals = species with ≥ 1 unpaired electron. Reactive Oxygen Species (ROS) include $O_2^{\bullet-}$, H_2O_2 , $OH^{\bullet}$. Harmful in excess but essential in controlled amounts.
2. **Beneficial role — Phagocytosis:** NADPH oxidase in neutrophils/macrophages generates $O_2^{\bullet-} \rightarrow$ dismuted to $H_2O_2 \rightarrow$ myeloperoxidase + $Cl^- \rightarrow HOCl$ (hypochlorous acid) $\rightarrow$ potent bactericidal. Essential for innate immunity. Deficiency (CGD) $\rightarrow$ recurrent life-threatening infections.
3. **Cell signalling:** H_2O_2 acts as second messenger — activates NF- κ B, MAP kinases, VEGF signalling. Low concentrations regulate insulin receptor signalling (inhibit PTP1B phosphatase $\rightarrow$ prolong insulin signal). Superoxide regulates vascular tone.
4. **Thyroid hormone synthesis:** DUOX (thyroid oxidase) generates H_2O_2 which is essential for thyroid peroxidase-catalysed iodination of thyroglobulin $\rightarrow$ T3 and T4 synthesis. No H_2O_2 = no thyroid hormone.
5. **Nitric oxide:** NO $^{\bullet}$ (a radical) = powerful physiological vasodilator (activates guanylyl cyclase $\rightarrow$ cGMP $\rightarrow$ smooth muscle relaxation). Also neurotransmitter and immune effector. Essential for penile erection (PDE5 inhibitors work on this pathway).
6. **Reproductive function:** ROS from sperm NADPH oxidase required for sperm capacitation and acrosome reaction $\rightarrow$ fertilisation. Excessive ROS (from leukocytes in semen) = male infertility.
7. **Examiner tip:** Balance is key — physiological ROS at low levels = beneficial; excess ROS (oxidative stress) = damage. Antioxidant defence systems (SOD, catalase, GPx, Vit E, C) maintain this balance.

 EQ – RENAL CLEARANCE AS PREDICTOR OF RENAL FAILURE [PYQ 2014, 2018]

"Renal clearance study is a nearly predictor of impending renal failure."

1. **Definition of clearance:** Volume of plasma cleared of a substance per unit time (mL/min). $CrCl = \frac{\text{Urine Cr} \times \text{Volume/time}}{\text{Plasma Cr}}$. Represents GFR in clinical practice.
2. **Ideal clearance marker:** Inulin clearance = true GFR (freely filtered, not secreted, not reabsorbed). For clinical use, creatinine clearance is the standard approximation.
3. **Why clearance detects failure early:** Serum creatinine remains in normal range until ~50% of nephron mass is lost (due to compensatory hyperfiltration in remaining nephrons). Creatinine clearance falls progressively as nephrons are lost — detects early renal impairment before serum Cr rises.
4. **CKD staging by GFR:** Stage 1 (≥ 90) $\rightarrow$ Stage 2 (60–89) $\rightarrow$ Stage 3a (45–59) $\rightarrow$ Stage 3b (30–44) $\rightarrow$ Stage 4 (15–29) $\rightarrow$ Stage 5/ESRD (<15 mL/min). Clearance study allows staging and guides timing of renal replacement therapy.
5. **Modern formulae:** MDRD (Modification of Diet in Renal Disease) and CKD-EPI equations estimate GFR from serum Cr, age, sex, race without 24-hour urine collection. Useful for population screening.
6. **Clinical implications:** Drug dosing adjustments required when GFR <60 (metformin, NSAIDs, antibiotics). Early nephrology referral when GFR <30 . Dialysis planning when GFR approaching 15.

EQ - GLUTATHIONE IS IMPORTANT MEDIATOR OF DETOXIFICATION [PYQ 2016-S, 2022]

"Glutathione is an important mediator for detoxification of toxic materials in humans."

1. **Structure:** Tripeptide — γ -Glu-Cys-Gly. The -SH (thiol) of cysteine is the active group. The unusual γ -glutamyl bond protects from non-specific peptidase digestion. Synthesised in virtually all cells; liver is major site.
2. **Antioxidant role:** GPx uses GSH to reduce $\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{GSSG}$. $\text{ROOH} \rightarrow \text{ROH} + \text{GSSG} + \text{H}_2\text{O}$. Regenerated by glutathione reductase (uses NADPH from HMP shunt). Maintains reducing environment in cells.
3. **Phase II detoxification (GST):** Glutathione S-transferase conjugates GSH with electrophilic toxins $\rightarrow$ glutathione conjugate $\rightarrow$ cysteinyl conjugate $\rightarrow$ N-acetylcysteine conjugate (mercapturic acid) $\rightarrow$ excreted in urine. Detoxifies NAPQI (paracetamol), epoxides (carcinogens), disulfiram metabolites.
4. **RBC protection:** RBCs produce NADPH only via HMP shunt (G6PD). G6PD deficiency $\rightarrow$ $\downarrow$ NADPH $\rightarrow$ $\downarrow$ GSH regeneration $\rightarrow$ oxidised Hb $\rightarrow$ Heinz bodies $\rightarrow$ haemolytic anaemia.
5. **Leukotriene synthesis:** $\text{LTC}_4 = \text{leukotriene A}_4 + \text{GSH}$ (LTC₄ synthase). LTC₄, LTD₄, LTE₄ (cysteinyl leukotrienes) = potent bronchoconstrictors in asthma. GSH is therefore also a precursor for inflammatory mediators.
6. **Clinical application:** N-Acetylcysteine (NAC) — antidote for paracetamol overdose (replenishes GSH). Also used in contrast nephropathy prevention. Oral NAC (Mucomyst) for CF (mucolytic — reduces disulphide bonds in mucus).

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Quickfire Revision Table



EXAM DAY - HIGH-YIELD MODULE 09

QUESTION	ANSWER
Henderson-Hasselbalch for bicarbonate buffer	$\text{pH} = 6.1 + \log [\text{HCO}_3^-] / [\text{H}_2\text{CO}_3]$. Normal ratio = 20:1 → pH 7.4
Normal pH / pCO₂ / HCO₃⁻	pH 7.35–7.45 / pCO ₂ 35–45 mmHg / HCO ₃ ⁻ 22–26 mEq/L
Most powerful acid-base regulator	Kidney (slowest but can completely restore pH). Lungs = fast but partial
Mnemonic for ABG interpretation	ROME = Respiratory Opposite, Metabolic Equal (pH and primary value directions)
Anion gap formula + normal	$\text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-) = 8\text{--}16 \text{ mEq/L}$
MUDPILES	Methanol, Uraemia, DKA, Paraldehyde, Isoniazid, Lactic acidosis, Ethylene glycol, Salicylates — all cause High AG metabolic acidosis
Paradoxical aciduria (persistent vomiting)	Metabolic alkalosis + hypokalemia → K ⁺ exits cells + H ⁺ enters → excess H ⁺ in tubule → acidic urine despite systemic alkalosis
Best cholestasis marker	ALP (Alkaline Phosphatase) + GGT
Best hepatocellular damage marker	ALT/SGPT (most liver-specific). AST:ALT >2 in alcoholic hepatitis
Dark urine + pale stool = ?	Obstructive jaundice (conjugated bili → urine; no bile → gut → pale stool)
Why phototherapy in neonatal jaundice	Blue light converts unconjugated bili → lumirubin/photobilirubin (water-soluble) → excreted without conjugation
Most specific RFT for GFR	Creatinine clearance (best clinical estimate). Inulin clearance = true GFR (not clinical)
Why Cr more specific than urea	Creatinine = constant rate, diet-independent. Urea = affected by protein intake, dehydration, GI bleeding, liver function
HbA1c reflects glucose over	2–3 months (RBC lifespan 120 days). Falsely low in haemolysis
Most reactive/damaging ROS	Hydroxyl radical (OH•) — no enzyme scavenger
SOD reaction	$2\text{O}_2^{\bullet -} + 2\text{H}^+ \rightarrow \text{H}_2\text{O}_2 + \text{O}_2$ (Cu/Zn-SOD in cytoplasm, Mn-SOD in mitochondria)
SOD mutation → disease	SOD1 mutation → Familial ALS
GPx cofactor (trace element)	Selenium (Se). Deficiency → Keshan disease (cardiomyopathy)
NAC antidote for	Paracetamol overdose — replenishes GSH to neutralise NAPQI
CYP2E1 activates	Paracetamol → NAPQI (toxic). Also activated by chronic alcohol, isoniazid
Most common Phase II reaction	Glucuronidation (UDP-glucuronic acid + UGT)
P450 absorbs light at	450 nm (when CO-bound) — hence the name CYP450
Rifampicin + OCP failure — why?	Rifampicin = CYP inducer → ↑ oestrogen metabolism → ↓ OCP efficacy → contraceptive failure
ELISA sensitivity type for HIV screening	Indirect ELISA — detects anti-HIV antibodies in patient serum

Lipid peroxidation marker

MDA (malondialdehyde) measured as TBARS (thiobarbituric acid reactive substances)

MODULE 09 MASTER MNEMONICS

● BLOOD BUFFERS

"Big Healthy People Play" = Bicarbonate, Haemoglobin, Protein, Phosphate

● ABG RULE

"ROME" = Respiratory Opposite (pH↓/pCO₂↑), Metabolic Equal (pH↓/HCO₃⁻↓)

● HIGH AG METABOLIC ACIDOSIS

"MUDPILES" = Methanol, Uraemia, DKA, Paraldehyde, Isoniazid, Lactic acidosis, Ethylene glycol, Salicylates

● PHASE II CONJUGATION

"GluSAGMe" = Glucuronidation, Sulphation, Acetylation, Glutathione, Methylation

MODULE 09 · CLINICAL BIOCHEMISTRY

Caffeine & Cadaver · MBBS Biochemistry Notes · 2010-2025

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